

Business Requirements Document (BRD)

Project: Cebuana Lhuillier Parking Reservation System

Version: 2.0

Date: February 24, 2026

Stakeholders: Corporate Facilities, IT, End Users (Employees), Parking Attendants

1. Executive Summary

Cebuana Lhuillier requires a corporate parking reservation system to manage limited parking spaces across multiple buildings. The system serves three user roles — End Users (employees/parkers), Administrators, and Parking Attendants — each with distinct workflows. The solution must support multi-building, multi-floor parking management with zone-based access control, VIP designations, and real-time occupancy tracking.

2. Business Objectives

ID	Objective	Success Metric
BO-1	Reduce parking conflicts and double-bookings	Zero double-booking incidents
BO-2	Streamline parking slot allocation across buildings	90%+ slot utilization rate
BO-3	Enable self-service booking for employees	80%+ adoption within 3 months
BO-4	Provide management visibility into parking usage	Real-time dashboards and reports
BO-5	Enforce corporate parking policies (zones, VIP)	100% policy compliance
BO-6	Track and reduce parking no-shows	No-show rate below 5%

3. Stakeholder Roles

Role	Description	Key Needs
End User (Parker)	Employee who books parking	Quick booking, vehicle management, mobile-friendly

Role	Description	Key Needs
Admin	Facilities/IT administrator	Full control, user management, reporting, configuration
Attendant	On-site parking staff	View daily reservations, confirm arrivals, report no-shows

4. User Stories

4.1 Authentication & Account Management

ID	User Story	Priority	Acceptance Criteria
US-AUTH-01	As an employee, I want to log in with my corporate email and password so that I can access the system securely.	P0	Login form with email/password, session cookie issued, redirect to role-appropriate dashboard
US-AUTH-02	As a new user, I want to be prompted to change my temporary password on first login so that my account is secure.	P0	<code>must_change_password</code> flag enforced, redirect to change-password screen
US-AUTH-03	As a user, I want my session to expire after 30 minutes of inactivity so that unauthorized access is prevented.	P1	JWT token expires after 30 minutes, automatic redirect to login
US-AUTH-04	As a user, I want to log out securely so that my session is terminated.	P0	Cookie cleared, session revoked in DB
US-AUTH-05	As an admin, I want to reset a user's password so that I can help employees who are locked out.	P1	Admin can set new password for any user
US-AUTH-06	As a user, I want to change my own password from my profile page by verifying my current password first, so that my account remains secure.	P1	Change password form with current/new/confirm fields; current password verified before update
US-AUTH-07	As a user who forgot my password, I want to submit my	P1	"Forgot Password" link on login page; admin notified via in-app notification;

ID	User Story	Priority	Acceptance Criteria
	email on the login page to notify an admin, so that I can regain access to my account.		email enumeration prevented (always returns success)

4.2 End User — Vehicle Management

ID	User Story	Priority	Acceptance Criteria
US-VEH-01	As a user, I want to register my vehicle(s) with plate number, make, model, and color so that I can book parking.	P0	Vehicle CRUD with all fields stored. Plate number must be unique across the system.
US-VEH-02	As a user, I want to manage multiple vehicles so that I can book with any of my cars.	P1	List, add, delete vehicles from profile
US-VEH-03	As a user, I want to see my vehicles listed on my dashboard for quick access.	P2	Dashboard shows vehicle cards

4.3 End User — Parking Reservation

ID	User Story	Priority	Acceptance Criteria
US-RES-01	As a user, I want to book a parking slot by selecting a building, floor, date(s), time range, and vehicle so that I can reserve my spot.	P0	Multi-step booking flow, slot confirmation
US-RES-02	As a user, I want to book multiple individual dates (not weekly) in a single booking so that I can plan ahead.	P0	Date picker allows multiple discrete dates
US-RES-03	As a user, I want to see which slots are available for my selected date and floor so that I can choose a spot.	P0	Real-time slot availability grid
US-RES-04	As a user, I want to receive a QR code for my reservation so that I can present it at the parking gate.	P1	QR code generated per reservation, viewable in app
US-RES-05	As a user, I want to cancel my reservation so that the slot is released for others.	P0	Cancel button on active reservations, slot immediately freed
		P1	

ID	User Story	Priority	Acceptance Criteria
US-RES-06	As a user, I want to view my active and past reservations in separate tabs so that I can track my history.		"Active" tab shows today+future, "History" tab shows past/cancelled
US-RES-07	As a user, I want my default booking times to auto-fill when I create a reservation so that booking is faster.	P2	User preferences applied, admin defaults as fallback
US-RES-08	As a user, I want to set my preferred default booking start/end times on my profile page.	P2	Profile page with time inputs, saved to user record
US-RES-09	As a user, I want to see how many buildings are in my zone on the dashboard so that I know my options.	P2	Dashboard "Buildings" card shows zone-filtered count
US-RES-10	As a user, I want to be notified about booking confirmations and no-show warnings so that I stay informed.	P1	In-app notification bell with unread count
US-RES-11	As a user, I want to see hourly slot availability on a visual timeline so that I can choose the best time to park.	P1	Hourly timeline (6AM–10PM) per slot, color-coded booked vs. available hours
US-RES-12	As a user, I want to join a waitlist when all slots are fully booked so that I am notified when a slot becomes available.	P1	Waitlist join, position tracking, in-app notification when slot opens, configurable booking window

4.4 End User — Zone & Building Rules

ID	User Story	Priority	Acceptance Criteria
US-ZONE-01	As a user, I want to book in my main building without restrictions so that my daily commute is simple.	P0	Booking in main building: no extra validation
US-ZONE-02	As a user, I want to book in any building within my assigned zone so that I have flexibility.	P0	Zone buildings visible and bookable
US-ZONE-03	As a user assigned to a zone, I want to book in a non-main building (within zone) for a single day with a reason so that external visits are tracked.	P1	Single-day restriction + reason field for non-main buildings

4.5 Admin — User Management

ID	User Story	Priority	Acceptance Criteria
US-ADM-01	As an admin, I want to create individual users with email, name, company, role, and building assignment.	P0	User creation form with all fields
US-ADM-02	As an admin, I want to bulk upload users via CSV/Excel with columns for name, email, company, role, main building, and zone.	P0	File upload, auto-resolve building/zone names, success/error summary
US-ADM-03	As an admin, I want to block/unblock users so that I can enforce access policies.	P1	Block/unblock toggle, blocked users cannot log in
US-ADM-04	As an admin, I want to assign a main building to a user so that their default booking location is set.	P0	Main building field on user edit form
US-ADM-05	As an admin, I want to tag users as VIP or Group Head so that they receive special privileges.	P1	Tags (vip, group_head) on user records
US-ADM-06	As an admin, I want to set default booking times per user so that corporate schedules are enforced.	P2	Start/end time fields in user edit
US-ADM-07	As an admin, I want to filter users by company, zone, main building, and role so that I can find users quickly.	P1	Filter dropdowns with "No Zone" / "No Building" options
US-ADM-08	As an admin, I want to select multiple users and bulk-assign them to a zone or set their main building so that onboarding is efficient.	P1	Multi-select checkboxes, bulk action bar
US-ADM-09	As an admin, I want to download a CSV template with example data including main_building and zone columns.	P2	Template download with sample rows
US-ADM-10	As an admin, I want to see when users were created and last updated so that I can track account activity.	P2	created_at and updated_at columns in user table
US-ADM-11	As an admin, I want to configure waitlist settings per building (enable/disable, notification window) so that I can manage parking demand.	P1	Waitlist toggle and window minutes in parking config
US-ADM-12	As an admin, I want to record and view a user's Job Family (e.g., Department Head) so that organizational context is captured.	P1	Job Family field on user profile, bulk upload CSV, admin user form

4.6 Admin — Building & Slot Management

ID	User Story	Priority	Acceptance Criteria
US-BLD-01	As an admin, I want to create buildings with name, address, and number of floors.	P0	Building creation form
US-BLD-02	As an admin, I want to add floors to a building and upload a floor layout image.	P0	Floor creation with image upload
US-BLD-03	As an admin, I want to add parking slots (alphanumeric labels) to a floor individually or in bulk.	P0	Single and batch slot creation
US-BLD-04	As an admin, I want to block/unblock individual parking slots (e.g., for maintenance) without deleting them.	P1	Slot status toggle (active/maintenance)
US-BLD-05	As an admin, I want to delete slots that are no longer in use.	P1	Slot deletion
US-BLD-06	As an admin, I want to view available slots filtered by building, floor, and date.	P0	Available slots API with filters

4.7 Admin — Zone Management

ID	User Story	Priority	Acceptance Criteria
US-ZN-01	As an admin, I want to create zones that group multiple buildings together.	P0	Zone CRUD with building assignment
US-ZN-02	As an admin, I want to assign users to zones so that their building access is controlled.	P0	User assignment within zone management
US-ZN-03	As an admin, I want to add/remove buildings from a zone.	P1	Building multi-select in zone editor

4.8 Admin — Parking Configuration

ID	User Story	Priority	Acceptance Criteria
US-CFG-01	As an admin, I want to set default parking hours (start/end time) per building.	P0	Parking config form per building
US-CFG-02	As an admin, I want to configure how many days in advance users can book.	P1	Booking window days setting

ID	User Story	Priority	Acceptance Criteria
US-CFG-03	As an admin, I want to set a release time (when slots become bookable each day).	P1	Release time field
US-CFG-04	As an admin, I want to enable/disable automatic no-show slot release.	P2	Toggle for auto-release with time offset
US-CFG-05	As an admin, I want to enable main building exclusivity for a building so that only assigned employees can park there.	P1	<code>main_building_exclusive</code> toggle in parking config

4.9 Admin — Reports & Analytics

ID	User Story	Priority	Acceptance Criteria
US-RPT-01	As an admin, I want to view a dashboard with total reservations, active users, occupancy rate, and building breakdown.	P0	Stats API with summary and per-building data
US-RPT-02	As an admin, I want to filter reports by date range and building.	P1	Date range and building filter on reports page
US-RPT-03	As an admin, I want to generate AI-powered insights from parking data so that I can identify trends.	P2	AI insights generation with markdown rendering
US-RPT-04	As an admin, I want to view a history of previously generated insights.	P2	Insights history tab with timestamps
US-RPT-05	As an admin, I want to view all reservations across buildings with search and filter.	P1	Admin reservations table with status filters

4.10 Admin — Event Blocking

ID	User Story	Priority	Acceptance Criteria
US-EVT-01	As an admin, I want to block multiple parking slots for a specific date and time range for an event, so that those slots are unavailable to regular users during the event.	P0	Visual slot map for selecting slots; blocked slots become confirmed reservations with <code>booking_type: event_block</code> ; slots appear as taken in the booking grid
US-EVT-02	As an admin, I want to name an event block and provide a reason,	P0	Event name and reason fields required

ID	User Story	Priority	Acceptance Criteria
	so that the purpose of the blocked slots is documented.		
US-EVT-03	As an admin, I want to view all active event blocks and delete them when no longer needed, so that slots are released back to users.	P1	Event block list with name, date, time, slot count; delete action removes all associated reservation records

4.11 Admin — Site Customization

ID	User Story	Priority	Acceptance Criteria
US-SITE-01	As an admin, I want to customize the login page heading, description, and badges so that the app reflects our brand.	P2	Site content editor in admin settings

4.12 Admin — Building Policies

ID	User Story	Priority	Acceptance Criteria
US-POL-01	As an admin, I want to create a dedicated-slot policy for a building so that certain slots are reserved for specific users and vehicles.	P0	Policy CRUD per building, policy_type field
US-POL-02	As an admin, I want to require parking stickers for a building so that only registered vehicles can park there.	P1	<code>requires_sticker</code> toggle on policy
US-POL-03	As an admin, I want to register users and their vehicles to specific slots so that assignments are enforced.	P0	Slot registration CRUD with user, vehicle_plate, sticker_number
US-POL-04	As an admin, I want to designate open floors within a policy building so that unregistered users can still park on those floors.	P1	<code>open_floor_ids</code> list on policy
US-POL-05	As an admin, I want the user selection dropdown to be filtered by the building's zone when assigning slot registrations.	P1	Zone-filtered user list
US-POL-06	As an admin, I want to select a vehicle from the user's registered vehicles	P1	Vehicle dropdown populated from user's vehicles

ID	User Story	Priority	Acceptance Criteria
	(dropdown) when assigning a slot registration.		

4.13 Parking Attendant

ID	User Story	Priority	Acceptance Criteria
US-ATT-01	As an attendant, I want to view all reservations for a specific date and building so that I can manage daily parking.	P0	Daily reservation list with building filter
US-ATT-02	As an attendant, I want to confirm a reservation when a car arrives so that occupancy is tracked.	P0	Confirm button, status changes to "confirmed"
US-ATT-03	As an attendant, I want to report a no-show for users who don't arrive by a certain time, only for today or past dates.	P0	No-show button (hidden for future dates), backend date validation
US-ATT-04	As an attendant, I want to quickly return to today's date from any viewed date.	P2	"Today" button in date navigator
US-ATT-05	As an attendant, I want to scan a QR code to verify a reservation.	P1	QR scan endpoint returns reservation details
US-ATT-06	As an attendant, I want to see summary cards (Reserved, Confirmed, No Shows) for the current day.	P1	Status summary cards on attendant dashboard

5. Business Rules

Rule ID	Rule	Enforcement
BR-01	A user must have at least one registered vehicle to book a parking slot.	Frontend validation + backend check
BR-02	A slot can only be booked by one user per date/time range — no double-booking.	Backend atomic check on slot+date+status
BR-03	Booking in a non-main building (within zone) is limited to single-day and requires a reason.	Backend validation on booking type

Rule ID	Rule	Enforcement
BR-04	Blocked users cannot log in or make reservations.	Auth check on login, booking endpoints
BR-05	No-show can only be reported for reservations on today or past dates. A <code>no_show_at</code> timestamp is recorded when the no-show is tagged.	Backend date validation, frontend button hidden for future, <code>no_show_at</code> stored
BR-06	Sessions expire after 30 minutes.	JWT expiration + cookie max-age
BR-07	VIP/Group Head users have booking priority (tag-based).	Tags stored on user, available for priority logic
BR-08	Slots in "maintenance" status cannot be booked.	Excluded from available slots query
BR-09	Advance booking is limited by building-specific booking window (default 7 days).	Config-driven validation
BR-10	Users must change their temporary password on first login.	<code>must_change_password</code> flag enforcement
BR-11	Waitlist notifications expire after a configurable window (default 15 minutes).	Background task checks <code>notified_at</code> + window; next user notified on expiry
BR-12	No-show auto-release timing is based on when the reservation is tagged as a no-show (<code>no_show_at</code>), not on the reservation end time.	<code>no_show_at</code> timestamp set on status change; background task checks elapsed time
BR-13	A vehicle plate number must be unique across the entire system. No two vehicles can share the same plate.	Backend uniqueness check on <code>POST /api/vehicles</code> before insert
BR-14	For buildings with a sticker policy, the specific vehicle used for booking must have an active slot registration in that building. User-level registration alone is not sufficient.	Backend checks <code>vehicle_plate</code> match in <code>slot_registrations</code> during reservation creation
BR-15	For non-open floors in policy buildings, the vehicle must be registered to the specific slot being booked, not just any slot in the building.	Backend checks <code>slot_id + user_id + vehicle_plate</code> in <code>slot_registrations</code>
BR-16	Admin-initiated cancellations require a mandatory cancellation reason. The paker receives an in-app notification	Backend requires <code>reason</code> field on admin cancel endpoint; creates notification for

Rule ID	Rule	Enforcement
	containing the reason. The waitlist is then triggered for the freed slot.	parker with reason; then triggers waitlist notification
BR-17	When main building exclusivity is enabled for a building, only users whose <code>main_building</code> matches can make reservations. Admins are exempt.	Backend checks <code>main_building_exclusive</code> config + user's <code>main_building</code>
BR-18	Event blocks create actual reservation records with <code>booking_type: "event_block"</code> to prevent double-booking.	Backend creates confirmed reservations per blocked slot
BR-19	Password changes require verification of the current password (except for forced first-login changes).	Backend verifies <code>current_password</code> hash match
BR-20	Forgot password requests notify all admin users. Email enumeration is prevented by always returning a success message.	Backend creates notification for each admin, same response regardless of email existence

6. Non-Functional Requirements

ID	Requirement	Target
NFR-01	Mobile-responsive UI	Works on 375px+ screens
NFR-02	Page load time	< 3 seconds
NFR-03	API response time	< 500ms for 95th percentile
NFR-04	Concurrent users	200+ simultaneous
NFR-05	Data security	HttpOnly cookies, bcrypt, CSP headers, password not exposed in DOM
NFR-06	Availability	99.9% uptime target
NFR-07	Browser support	Chrome, Firefox, Safari, Edge (latest 2 versions)

7. Assumptions & Constraints

Assumptions

- Users are pre-provisioned by admin (no self-registration)
- All buildings and zones are configured by admin before users book
- Internet connectivity is available at all parking locations

Constraints

- MongoDB as the sole database (no SQL)
- Single-tenant deployment per corporate group
- AI insights require Emergent LLM API key with balance

Next: [System Requirements Specification](#)